Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/01/2004
PI Name:	Theofanous, Theofanis G. Ph.D.		
Project Title:	The Limits of Coolability in Thermal Systems for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
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No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dinh, Truc-Nam (University of California Santa Barbara)		
Grant/Contract No.:	NAG3-2761		
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Task Description:	This project is an integrative effort addressing practical aspects of boiling in the HEDS context. The work is a continuation and extension of a project that addressed coolability limits in pool boiling carried out over the previous funding cycle. A key result of this previous work was the identification of a scales separation phenomenon, that effectively "isolates" the microhydrodynamics on the heater surface from the chaotic two-phase flow motions away from it. This separation is effected by a persistent very high void fraction region (like a vapor "blanket"), and it is of immense significance in allowing to focus the physics of a long-standing seemingly intractable problem. Experiments and computations in this project will be used to examine applicability of scales separation as an organizing principle in convective boiling and boiling in fractional gravities. 1. Develop a basic understanding for the fine structure of the extended microlayer, as revealed through our "scale separation" idea and the dynamic thermal patterns on a nanofilm heater. 2. Define appropriate, measurable surface properties relevant to the rupture of microlayer, and relate those (quantitatively) to fluid properties and the rupture condition (the critical heat flux). 3. Determine the extent of applicability of scales separation in reduced gravity; that is, the point where CHF becomes water-supply-limited. 4. Determine the infleunce of convection on the results of Objetives 1 and 2. 5. Translate the results of all previous objectives to terms usable for predicting and enhancing CHF, as well as for use in a two-phase CFD framework. This project will provide a basis to develop a physics-based understanding of Boiling Crisis phenomenon in pool boiling (horizontal and vertical flat plates), and weakly-convecting flow boiling (vertical channel flow). We can then translate this understanding to a means for quantitative prediction of critical heat flux, as a function of local (near-wall) two-phase flow conditions, appropriate and convenient for interrogation (modelling) in the frame of detailed multiphase CFD simulations. Furthermore, the thus-gained understanding and experience can be used to enhance the coolability limits in both pool and forced convection flows by manipulating surface characteristics, and perhaps even flow geometries.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has been leveraged with a contract from the Department of Energy concerning the coolability limits of nuclear reactor vessels that in case of severe accidents are immersed in water pools by flooding the outside area with water. In turn the results of this fundamentally oriented work had a tremendous influence and impact on the large scale experiments we run for this nuclear application, and helped resolve this practical issue even though we are still in the process of developing a basic understanding.
Task Progress:	The purpose of this work is to understand the basic reason(s) that there is an upper limit on the level of power that can be removed from a heat-generating body by boiling of a liquid coolant. Such understanding is needed for reliable prediction, especially for systems as in space applications, for which a trial and error approach is not an option. Such understanding is also essential for realizing the full cooling potential of boiling, and thus opening up the way for innovative power production and conversion systems, as well as for addressing the cooling needs of new kinds of scientific equipment and instrumentation involving extremely intense heat sources (microelectronics, lasers, etc.). A wide range of terrestrial, aviation, and space technologies stand to benefit from such advancements. The symptoms of exceeding the cooling capacity of boiling are quite straightforward. The boiling surface dries out, and the temperature rises rapidly, thus leading to melting and destruction (hence the name “burnout”). Our problem is to find out why, in the presence of plentiful liquid around, the surface dries out. Research in the past half century led to the general concept, and belief, that the basic mechanism is insufficient liquid supply to the surface, this deficiency in supply being due to the repulsive effect of vapor generated at, and leaving the surface. This so-call hydrodynamic origin of burnout has led to several conceptual models that were translated to equations that purport to interpret experimental data. On the other hand, it is quite clear that understanding is still lacking, that predictions and control are not reliable, and that engineering system designs still rely on empirical correlations and trial and error approaches. In the first phase of our work under NASA support we demonstrated that in pool boiling the hydrodynamics set only the stage in a broad sense, rather than affecting coolability through some criticality condition as envisioned previously. This radical departure from the past is manifest in terms of the “scales separation phenomenon” discovered in this work. In the present, second phase, of the project the specific goal is to understand scales separation with all its implications for pool boiling, and to find out the extent to which it dominates behavior in convective (flow) boiling. Simply expressed, scales separation is due to a thin vapor layer that blankets the heater surface, while a micron-scale liquid film continues to provide cooling by evaporation. Dryout originates on this film as micron-scale spots, that spread and eventually merge to cause burnout. Prior to the inception of such an unstable condition, the thin film remains well organized, and its uniformity of behavior is in stark contrast to the chaotic liquid motions above it (above the vapor blanket). We say the heater-microlayer system is autonomous, while any excess liquid that reaches it, is rejected by the action of vapor production. In this way we have transformed and focused our problem (of coolability limits) to the stability of dry spots that form at the points at which vapor bubbles are nucleated on the heater surface. This in turn lead to hypotheses, and experiments that confirmed the importance of nanomorphology, and chemical constitution of the surface, as well as of the chemical composition of the liquid (including nanoparticles and impurities) being boiled, in affecting this onset of instability, and thus the power level needed for burnout. These experimental results in turn led to theoretical concepts involving microhydrodynamic behavior, such as molecular wicking (a mechanism for enhancing the spread of liquid molecules towards a dried-out region of the heating surface), and of vapor recoil (a mechanism that resists spreading). Still taking advantage of the scales separation phenomenon we are now working with the Microlayer itself, in a spray apparatus that allows independent control of thickness and detailed, high resolution observations from above (high speed video) as well as from bellow (high speed infrared visualization). In parallel we developed a forced convection apparatus that can conveniently accommodate the heater test sections from our pool boiling experiments, and we are entering initial testing in this too. This work has been leveraged with a contract from the Department of Energy concerning the coolability limits of

	nuclear reactor vessels that in case of severe accidents are immersed in water pools by flooding the outside area with water. In turn the results of this fundamentally oriented work had a tremendous influence and impact on the large scale experiments we run for this nuclear application, and helped resolve this practical issue even though we are still in the process of developing a basic understanding.
Bibliography Type:	Description: (Last Updated: 10/17/2007)
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